

National Academy of Science and Technology

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Pre-University Examination

Level: Bachelor

Semester – Fall

Year : 2025

Programme: BE

Full Marks : 100

Course: Digital Signal Analysis Processing

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary-coded words representing samples of an input signal $X_a(t) = 5 \cos 800\pi t + 3 \cos 1200\pi t$. The link is operated at 20,000 bits/s and each input sample is quantized into 1024 different voltage levels. 8
 - i) What is the sampling frequency and the folding frequency?
 - ii) What is the Nyquist rate for the signal $X_a(t)$?
 - iii) What are different frequencies in the resulting discrete-time signals $x[n]$?
 - iv) What is the resolution Δ ? 7
- b) Highlighting the key features of Digital Signal processing, briefly differentiate Digital Signal Processor with Analog Signal Processor.
2. a) Define LTI system. Mention and explain the properties of LTI system. 8

OR 7

Show that the multiplication of two DFT sequences results in circular convolution

- b) Obtain a linear convolution of the following two discrete-time signals:
 $x(n) = \{0, 1, 2, 3\}$ and $h(n) = 2n[u(n) - u(n-3)]$
- 3. a) For the given system $y[n] = 0.7 y[n-1] - 0.1 y[n-2] + x[n]$, plot the pole zero diagram and magnitude response of the system 8
- b) Determine the causal signal $x[n]$ having Z-Transform $X(z) = 1/\{(1-2z^{-1})(1-z^{-1})^2\}$. 7
- 4. a) Define DFT. Also compute 4-point DFT of the following sequence: 7
 $x(n) = \{1, 2, 3, 4\}$
- b) Find 8-point DFT of the sequence $\{0, 0, 0, 0, 1, 1, 1, 1\}$ using radix 2 DITFFT. 8
- 5. a) If there is given a three stage lattice filter with coefficient $K_1 = 1/4$, 8

$K_2 = 1/2$, and $K_3 = 1/3$, determine the FIR filter coefficient for the direct-form structure. 7

- b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure.

$$H(Z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}.$$

6. a) Design an IIR filter by using Bi-linear transformation to meet the following specification $w_p = 0.25\pi$, $w_s = 0.55\pi$, $\alpha_s = 15\text{dB}$, $\alpha_p = 0.5\text{dB}$. 8

OR

Design a linear FIR filter using Kaiser window to meet the following

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi$$

$$\leq |H(e^{j\omega})| \leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi$$

- b) Draw the flowchart of Remez algorithm and explain it

OR

The transfer function of Analog filter is $H_a(s) = \frac{2}{(s+2)(s+4)}$ with

$T = 0.1$ sec. Design the IIR digital filter by using impulse invariance method.

7. Write short notes on (Any Two):

5×2

- a) Gibb's Phenomena
- b) Cascade structure for FIR system
- c) ROC of Z transform